3 Processes, techniques and specialist tools

3.5 Application, advantages and disadvantages of the following finishing techniques and methods of preservation in order to discriminate between them and select appropriately for use, including for the prevention of degradation:

Aim:

3.5(a) Understand the need for finishes – finishes – paints, varnishes, sealants, preservatives, anodising, electroplating, powder coating, oil coating, galvanisation, cathodic protection (including use of specialist tools)

Research

Task 1:

Research the ***'surface treatments'*** listed below and produce a report. Illustrate your answers with *diagrams or images* were possible.

Your report should give examples of the materials and products which are ***'protected'*** by these ***'surface finishes'***, *how they are applied and the advantages and disadvantages of each*. Complete the Google Docs worksheet.

1.Painted Coatings
2.Varnishing
3.Oil Coating
4.Powder Coating (Dip Coating - we can do this in school)
5.Anodising
6.Electroplating
7.Galvanising
8.Catholic Protection

Task 2: answer the additional questions.

1. Painted Coatings:

2. Varnishing:

3. Oil Coating:

4.Powder Coating (Dip Coating *- we can do this in school*):

5.Anodising:

6.Electroplating:

7.Galvanising:

8.Catholic Protection:

Task 2: Test Yourself:

1. State ***two*** methods used to apply paint to a product.
2. Describe the process of ***powder coating***.
3. State ***four*** metals that can be coated onto steel by ***electroplating***.
4. Name ***three*** steel products that are ***galvanised***.
5. Give ***two*** reasons for ***anodising*** an aluminium product.
6. Name the *material* used in ***anodising*** and a ***product*** that has an ***anodised finish***.
7. Suggest a *suitable finish* for *washing machine body panels*.
8. What ***sacrificial metal*** is used in the ***galvanising*** process?